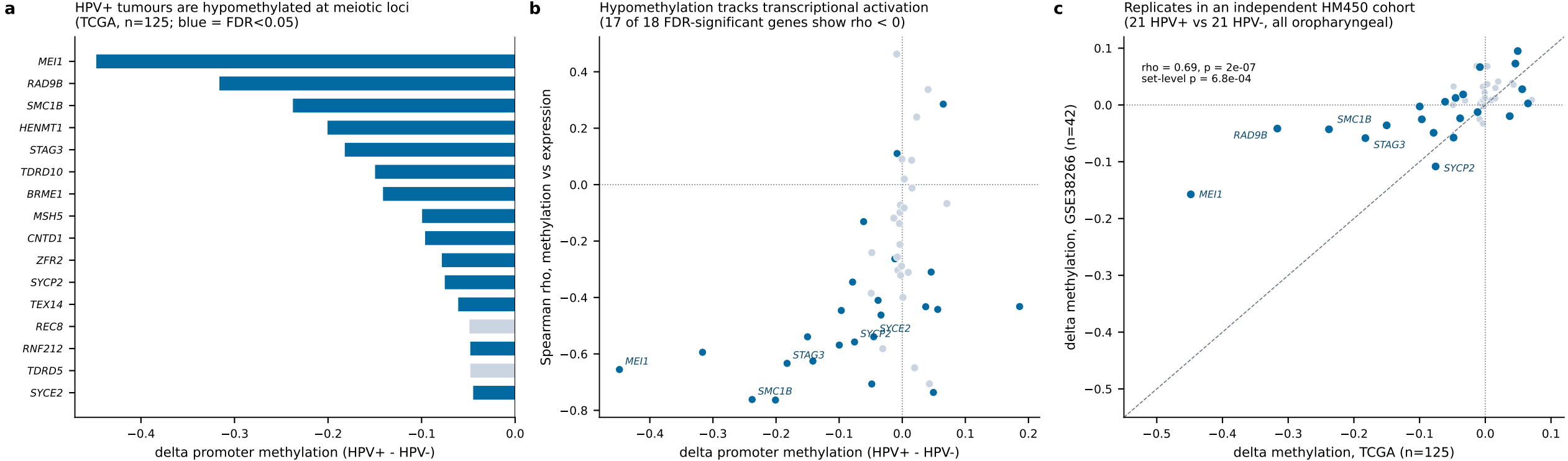
a
HPV+ tumours are hypomethylated at meiotic loci
(TCGA, n=125; blue = FDR<0.05)
MEI1
RAD9B
SMC1B
HENMT1
STAG3
TDRD10
BRME1
MSH5
CNTD1
ZFR2
SYCP2
TEX14
REC8
RNF212
TDRD5
SYCE2
−0.4
−0.3
−0.2
−0.1
0.0
delta promoter methylation (HPV+ - HPV-)
b
Hypomethylation tracks transcriptional activation
(17 of 18 FDR-significant genes show rho < 0)
Spearman rho, methylation vs expression
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
MEI1
SMC1B
STAG3
SYCP2
SYCE2
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
delta promoter methylation (HPV+ - HPV-)
c
Replicates in an independent HM450 cohort
(21 HPV+ vs 21 HPV-, all oropharyngeal)
rho = 0.69, p = 2e-07
set-level p = 6.8e-04
delta methylation, GSE38266 (n=42)
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
RAD9B
SMC1B
STAG3
SYCP2
MEI1
−0.5
−0.4
−0.3
−0.2
−0.1
0.0
0.1
delta methylation, TCGA (n=125)